

# SWITCHES IN PCBA

Types, Assembly & Integration Guide

Tactile • Slide • DIP • Toggle • Rotary • Membrane  
Push Button • Rocker • SMD vs. Through-Hole • Process Standards

PCBA Manufacturing & Switch Integration Standards

IPC/J-STD-001 | IPC-A-610 | IEC 61058 | UL 1054

# 1. SWITCHES IN PCBA OVERVIEW

**Switches** are electromechanical components that make or break electrical circuits by physically moving contacts. They are essential for user input, mode selection, configuration settings, and safety interlocks in virtually every electronic device. In PCBA assembly, switches range from microscopic SMD tactile switches to large panel-mounted toggle switches.

## Why Switches are Critical in PCBA:

- **User interface:** Power on/off, reset, mode selection, volume control
- **Configuration:** DIP switches for address/setting selection without software
- **Safety interlocks:** Emergency stop, door switches, lid detection
- **Test and programming:** Boot mode selection, factory test points
- **Signal routing:** Multiposition rotary switches for channel selection

## Switch Assembly Challenges:

- **Mechanical stress:** User-actuated switches endure repeated force — solder joints must be robust
- **Through-hole vs. SMD:** Different assembly processes and reliability profiles
- **Orientation:** Many switches are polarized — wrong placement = non-functional
- **Height/profile:** Switch actuator height affects enclosure design and user access
- **Environmental sealing:** IP-rated switches require gasket/seal integrity
- **Electrical rating:** Switch must handle voltage/current of application

■ **In-House Note:** Switches are often the most mechanically stressed components on a PCBA. While semiconductors fail from electrical overstress, switches fail from mechanical wear. Always verify switch cycle life (number of operations) matches product lifetime requirements.

## 2. TACTILE SWITCHES (PUSH BUTTON)

### 2.1 Overview

**Tactile switches** (also called tact switches or momentary push buttons) are the most common switch type in modern electronics. They provide momentary contact closure when pressed and open when released. The "tactile" name comes from the distinct "click" feedback felt when actuated.

### Key Characteristics:

- **Contact type:** Momentary (normally open) — closes only while pressed
- **Actuation force:** 1.0–3.5 N (100–350 gf) typical
- **Travel distance:** 0.2–0.5 mm (short travel) or 0.5–1.5 mm (long travel)
- **Electrical rating:** 12V DC, 50 mA typical (signal switching, not power)
- **Cycle life:** 100,000–1,000,000 operations
- **Package:** SMD (surface mount) or THT (through-hole)

### 2.2 Tactile Switch Types

|                               | Tactile Switches                                                                                      |                                                                                                                     |                                                                                                            |
|-------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| PCB Insert Hole Terminals     | <br>Standard Types | <br>Horizontal Pushbutton Types | <br>Hinge Lever Types |
| PCB Surface Mounted Terminals | <br>Standard Types | <br>Ultra Subminiature Types    | <br>Long-stroke Types |

|                               | Sealed Tactile Switches                                                                               |                                                                                                                |                                                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| PCB Insert Hole Terminals     | <br>Standard Types | <br>Radial Taping Packaging |                                                                                                              |
| PCB Surface Mounted Terminals | <br>Standard Types | <br>Low-profile Types      | <br>Medium-stroke Types |

Figure 1: Tactile Switch Types — PCB Insert Hole (THT) and Surface Mounted (SMD) Variants

### Standard Types:

- **4-pin square SMD:** 3×4 mm, 4×4 mm, 6×6 mm — most common for consumer electronics
- **2-pin rectangular SMD:** 2×3 mm, 2×4 mm — compact applications
- **Through-hole (THT):** 6×6 mm with 4 leads — stronger mechanical attachment
- **Side-actuated:** Push button on edge of board — space-saving design
- **Hinge lever:** Extended actuator for easier access — industrial applications
- **Sealed/waterproof:** IP67 rated with rubber boot — outdoor equipment

## 2.3 Assembly Process

### SMD Tactile Switches:

- **Stencil:** Standard 150 µm stencil; 1:1 aperture to pad size
- **Paste:** Type III SAC305 no-clean
- **Placement:** Standard pick-and-place; verify orientation marking
- **Reflow:** Standard SAC305 profile; peak 245–250°C
- **Inspection:** AOI for presence and orientation; verify no solder bridging

### THT Tactile Switches:

- **Insertion:** Manual or automated axial insertion machine
- **Soldering:** Wave solder or selective solder (NOT reflow — plastic melts)
- **Mechanical strength:** Leads provide superior mechanical attachment vs. SMD
- **Inspection:** Visual for solder fillet; pull test for mechanical integrity

## 2.4 Common Defects

- **Bridging:** Solder bridges between adjacent pins — reduce paste volume
- **Insufficient solder:** Weak joint prone to cracking under actuation force
- **Wrong orientation:** Switch pins misaligned with PCB pads
- **Actuator damage:** High placement force or improper handling cracks button
- **Flux residue:** No-clean residue under switch affects tactile feel

■ **Caution:** Tactile switches are plastic-bodied. Do NOT expose to wave solder or reflow temperatures beyond specification. Some low-cost switches have maximum temperature of 245°C — verify datasheet before reflow.

## 3. SLIDE SWITCHES

### 3.1 Overview

**Slide switches** (also called toggle or SPDT switches) use a sliding actuator to move contacts between positions. They maintain their state (latching) unlike momentary tactile switches. Common configurations include SPST (single pole, single throw), SPDT (single pole, double throw), and DPST (double pole, single throw).

#### Key Characteristics:

- **Contact type:** Maintained (latching) — stays in position until moved
- **Positions:** 2-position (ON-ON or ON-OFF) or 3-position (ON-OFF-ON)
- **Electrical rating:** 0.3A–3A at 12–30V DC typical
- **Cycle life:** 10,000–50,000 operations
- **Package:** SMD, THT, or side-mount (right-angle)

### 3.2 Slide Switch Types

#### MSS-22D18G2 Mini Slide Switch 6/3Pin SMD DIP Side

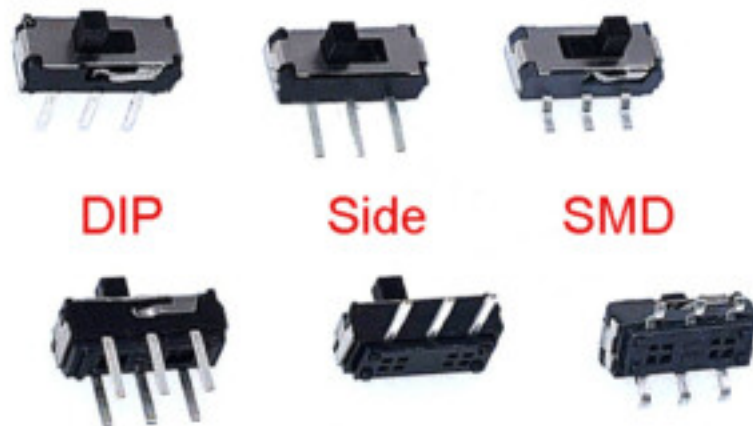


Figure 2: Mini Slide Switch Types — DIP, Side, and SMD Mounting Variants

#### Mounting Types:

- **DIP (Dual Inline Package):** Through-hole pins; strongest mechanical attachment
- **SMD (Surface Mount Device):** Gull-wing or J-lead terminals; compact
- **Side-mount:** Right-angle terminals; switch body parallel to PCB edge
- **Vertical:** Terminals on bottom; switch body perpendicular to PCB

### Circuit Configurations:

- **SPST:** Single Pole, Single Throw — simple ON/OFF
- **SPDT:** Single Pole, Double Throw — selects between two circuits
- **DPST:** Double Pole, Single Throw — two circuits ON/OFF simultaneously
- **DPDT:** Double Pole, Double Throw — two circuits select between two paths

## 3.3 Assembly Process

### SMD Slide Switches:

- **Stencil:** 1:1 aperture; watch for paste volume on gull-wing leads
- **Placement:** Verify actuator direction matches PCB silkscreen
- **Reflow:** Standard profile; ensure all terminals wet properly
- **Inspection:** AOI for coplanarity and solder fillet on all pins

### THT Slide Switches:

- **Insertion:** Manual or machine; ensure pins straight before insertion
- **Soldering:** Wave or selective solder
- **Mechanical support:** Some switches have mounting tabs — solder or screw
- **Actuator test:** Verify smooth sliding after assembly

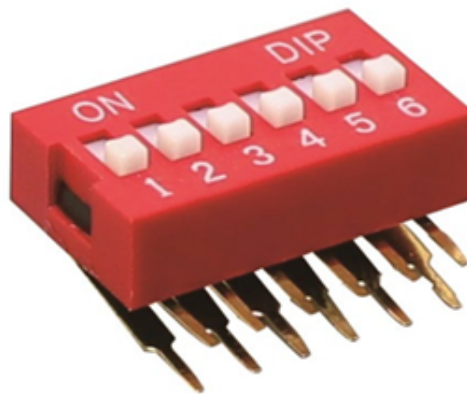
## 4. DIP SWITCHES

### 4.1 Overview

**DIP (Dual Inline Package) switches** are arrays of miniature slide switches in a standard IC package format. They are used for configuration settings, address selection, and feature enable/disable without requiring software programming.

#### Key Characteristics:

- **Pole count:** 1–12 switches in a single package (common: 2, 4, 6, 8, 10)
- **Pitch:** 2.54 mm (0.1 inch) — standard DIP IC spacing
- **Contact type:** SPST — each switch is independent ON/OFF
- **Electrical rating:** 24V DC, 25 mA typical (signal level)
- **Cycle life:** 1,000–2,000 operations per switch
- **Package:** Through-hole only (SMD variants exist but are rare)



*Figure 3: DIP Switch — 6-Position Through-Hole Configuration Switch*

### 4.2 Assembly Process

- **Insertion:** Manual or automated axial insertion
- **Orientation:** Pin 1 marking (dot or notch) aligns with PCB silkscreen
- **Soldering:** Wave solder or selective solder
- **Height:** Standard profile or low-profile (5.0 mm vs. 3.5 mm height)

- **Inspection:** Visual for solder fillet; verify all pins soldered
- **Functional test:** Verify continuity ON/OFF for each position

### 4.3 Common Applications

- **Device address:** I2C address selection (e.g., 0x20–0x27 for MCP23017)
- **Boot mode:** Select firmware update or normal boot
- **Feature enable:** Turn on/off optional features without code changes
- **Calibration:** Select gain or range settings
- **Test mode:** Enable factory test routines

### 4.4 SMD DIP Switch Alternative

For space-constrained designs, **SMD DIP switches** are available in compact packages (4×4 mm per switch). These are soldered during reflow but can be actuated with a small screwdriver or pick.

- **Package:** 4×4 mm or 3×3 mm per switch position
- **Actuation:** Small slider or recessed actuator
- **Reflow:** Standard SAC305 profile
- **Limitation:** Difficult to actuate after enclosure assembly



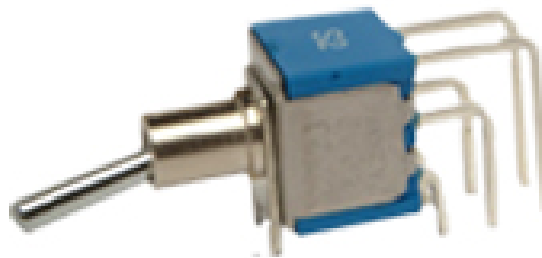
## 5. TOGGLE & ROCKER SWITCHES

### 5.1 Toggle Switches

**Toggle switches** use a lever (handle) that is flipped up/down or left/right to change contact state. They are robust, high-current switches used in industrial, automotive, and power applications.

#### Key Characteristics:

- **Contact type:** Maintained (latching) or momentary (spring-return)
- **Positions:** 2-position or 3-position (center-off)
- **Electrical rating:** 1A–20A at 125–250V AC/DC
- **Terminals:** Solder lug, quick-connect (spade), or screw terminal
- **Mounting:** Panel mount with threaded bushing; NOT PCB-mounted in most cases
- **Cycle life:** 10,000–100,000 operations



*Figure 4: Toggle Switch — PCB Mounting Type Vertical Operation*

#### PCB-Mounted Toggle Switches:

Some toggle switches are designed for direct PCB mounting with through-hole terminals instead of panel mounting. These are used when the PCB is accessible through an enclosure cutout.

- **Terminals:** Through-hole pins soldered to PCB
- **Mechanical support:** Threaded bushing passes through PCB and secures with nut
- **Orientation:** Lever direction must match enclosure cutout
- **Soldering:** Hand solder or wave solder; high thermal mass requires longer dwell

### 5.2 Rocker Switches

**Rocker switches** use a seesaw-style actuator that rocks between ON and OFF positions. They are common in consumer electronics, power strips, and appliance control panels.

### Key Characteristics:

- **Contact type:** Maintained (latching)
- **Positions:** 2-position (ON-OFF) or 3-position (ON-OFF-ON)
- **Electrical rating:** 6A–16A at 125–250V AC
- **Illuminated:** Many include LED indicator integrated into rocker
- **Mounting:** Panel mount or snap-in to enclosure; rarely PCB-mounted
- **Cycle life:** 10,000–50,000 operations

### PCB-Mounted Rocker Switches:

- **Small signal types:** 10×15 mm body with through-hole pins
- **Power types:** Larger body with solder lugs or quick-connect terminals
- **Illuminated:** Additional LED pins require correct polarity
- **Soldering:** Hand solder; verify terminal does not overheat plastic body

## 6. ROTARY SWITCHES & ENCODERS

### 6.1 Rotary Switches

**Rotary switches** use a rotating shaft to select one of multiple positions. They are used for channel selection, range setting, and multi-position configuration.

#### Key Characteristics:

- **Positions:** 2–12 positions (common: 3, 4, 6, 8, 10, 12)
- **Poles:** 1–4 poles (simultaneous switching of multiple circuits)
- **Contact type:** Break-before-make (BBM) or make-before-break (MBB)
- **Electrical rating:** 0.1A–5A at 30–250V AC/DC
- **Terminals:** Through-hole pins or solder lugs
- **Mounting:** Panel mount with threaded bushing; PCB support pins

### 6.2 Rotary Encoders

**Rotary encoders** are digital switches that convert rotational motion into electrical pulses. Unlike rotary switches, they have no fixed positions and can rotate continuously. They are used for volume control, menu navigation, and position feedback.

#### Types:

- **Mechanical (contact) encoder:** Physical contacts generate quadrature A/B pulses
- **Optical encoder:** LED and photodetector read coded disk — no contact wear
- **Magnetic encoder:** Hall effect sensors detect magnetic field rotation
- **Incremental:** Outputs relative position (quadrature A/B)
- **Absolute:** Outputs unique code for each position (parallel or serial)



Figure 5: PCB-Mounted Rotary Encoder Switches — Various Pin Configurations

### Encoder with Push Button:

Many rotary encoders include an integrated **push button** (tactile switch) activated by pressing the shaft. This provides "click to select" functionality for user interfaces.

- **Terminals:** 5 pins total — 2 for encoder A/B, 1 for common, 2 for push button
- **Mounting:** Through-hole with threaded bushing and nut
- **Soldering:** Hand solder; support pins take mechanical load
- **Application:** Volume knobs, scroll wheels, menu selectors

### 6.3 Assembly Process

- **Insertion:** Manual; align flat on shaft with knob set screw
- **Orientation:** Pin 1 (common) must align with PCB silkscreen
- **Mechanical support:** Tighten bushing nut to 0.5–1.0 N·m
- **Soldering:** Hand solder all pins; verify no cold joints
- **Test:** Verify pulse sequence (A leads B for CW rotation)

## 7. MEMBRANE SWITCHES

### 7.1 Overview

**Membrane switches** are low-profile, flexible input devices consisting of multiple layers of polyester or polycarbonate films with printed conductive circuits. They are used in appliances, medical devices, industrial controls, and consumer electronics where a flat, sealable surface is required.

#### Construction Layers:

- **Graphic overlay:** Top layer with printed labels and colors
- **Spacer layer:** Adhesive with cutouts for button cavities
- **Circuit layer:** Silver or carbon conductive traces on polyester
- **Rear adhesive:** Mounts to PCB or enclosure backplate
- **Domes (optional):** Metal or polyester snap domes for tactile feedback



Figure 6: PCB Membrane Switch — Graphic Overlay (Left) and PCB Circuit (Right)

### 7.2 Types of Membrane Switches

#### Flat (Non-Tactile):

- No physical feedback — user cannot feel actuation

- Lowest cost; longest life (1,000,000+ operations)
- Used in microwave ovens, remote controls

### Tactile (Metal Dome):

- Stainless steel dome provides snap feedback
- Audible click and tactile feel
- 500,000–1,000,000 operations
- Used in medical devices, industrial panels

### PCB-Based Membrane:

- Membrane overlay mounted directly on PCB with exposed pads
- PCB contains gold-plated or carbon contact pads
- Connector: ZIF (zero insertion force) or soldered tail
- Used in high-reliability applications

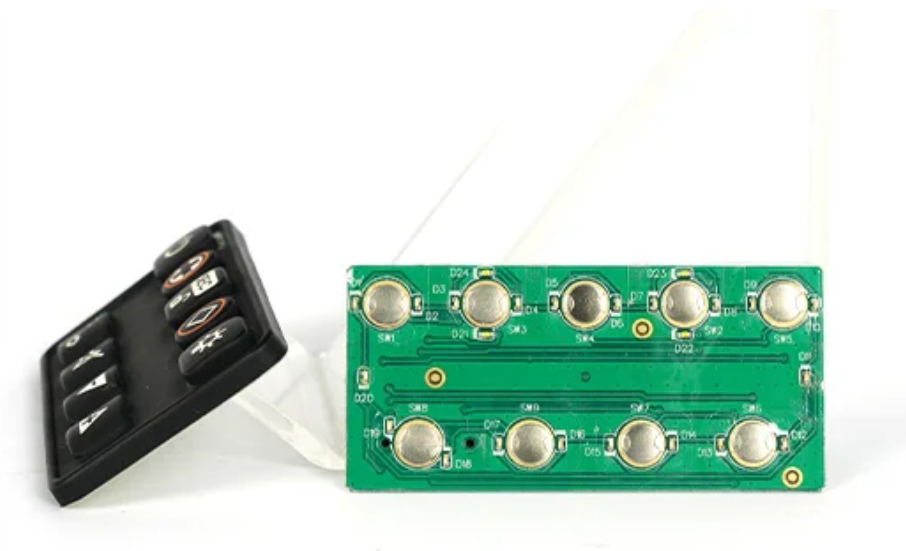


Figure 7: PCB Membrane Switch Assembly — Connector and Dome Array

## 7.3 Assembly Process

- **PCB preparation:** Clean and verify pad flatness; no solder mask on contact area
- **Dome placement:** Automated or manual placement of metal domes on PCB pads
- **Spacer application:** Adhesive spacer with button cutouts
- **Overlay lamination:** Graphic overlay aligned and pressed onto spacer
- **Connector attachment:** ZIF connector or soldered flex tail
- **Test:** Verify all buttons; check resistance ( $< 100 \, \Omega$  closed,  $> 10 \, \text{M}\Omega$  open)

## 7.4 Advantages & Limitations

### Advantages:

- Very low profile (0.5–1.5 mm total thickness)
- Sealed against moisture and dust (IP65–IP67 possible)
- Customizable graphics and colors
- Low cost at volume
- Easy to clean and disinfect (medical applications)

### **Limitations:**

- Limited tactile feedback (unless metal domes used)
- Not suitable for high-current switching
- Limited cycle life compared to mechanical switches
- Requires custom tooling for each design

## 8. SWITCH COMPLETE MATRIX

| Switch Type    | Mounting         | Actuation  | Rating   | Life     | Best For                 |
|----------------|------------------|------------|----------|----------|--------------------------|
| Tactile (SMD)  | SMD              | Momentary  | 12V/50mA | 100K–1M  | Consumer electronics     |
| Tactile (THT)  | THT              | Momentary  | 12V/50mA | 100K–1M  | Industrial, robust       |
| Slide (SMD)    | SMD              | Maintained | 30V/0.3A | 10K–50K  | Compact mode select      |
| Slide (THT)    | THT              | Maintained | 30V/3A   | 10K–50K  | Power routing            |
| DIP Switch     | THT              | Maintained | 24V/25mA | 1K–2K    | Configuration, address   |
| Toggle (Panel) | Panel            | Maintained | 250V/20A | 10K–100K | Power, industrial        |
| Toggle (PCB)   | THT              | Maintained | 30V/5A   | 10K–50K  | PCB power control        |
| Rocker (Panel) | Panel            | Maintained | 250V/16A | 10K–50K  | Appliances, power strips |
| Rotary Switch  | Panel/THT        | Maintained | 250V/5A  | 10K–50K  | Channel select, range    |
| Rotary Encoder | THT              | Continuous | 5V/10mA  | 100K–1M  | Volume, navigation       |
| Membrane       | Adhesive/P<br>CB | Momentary  | 24V/50mA | 500K–1M  | Medical, industrial UI   |

Table 1: Switch Types Complete Comparison Matrix



## 9. SWITCH SOLDERING & PROCESS STANDARDS

### 9.1 SMD Switch Reflow

SMD switches (tactile, slide, small rocker) are reflow-soldered like standard SMD components. However, their plastic bodies require temperature control.

- **Stencil:** 150 µm standard; 1:1 aperture to pad
- **Paste:** SAC305 Type III no-clean
- **Profile:** Standard SAC305; verify switch max temp (< 260°C peak)
- **Placement:** Standard P&P; verify orientation marking
- **Inspection:** AOI for presence, orientation, solder fillet
- **Cleaning:** No-clean flux acceptable; water-soluble requires DI wash

### 9.2 THT Switch Soldering

Through-hole switches require wave, selective, or hand soldering. The high mechanical stress on switch terminals demands robust solder joints.

#### Wave Soldering:

- **Flux:** RF-800 (no-clean) or WS715M (water-soluble)
- **Pot temp:** 250–260°C SAC305; 240–250°C SnPb
- **Contact time:** 3–5 seconds
- **Preheat:** 100–130°C top side
- **Inspection:** Visual for 360° wetting; minimum 75% hole fill

#### Hand Soldering:

- **Temperature:** 350–400°C for large terminals
- **Dwell:** 3–5 seconds per terminal
- **Technique:** Heat terminal, feed solder from opposite side
- **Warning:** Excessive heat melts plastic body — use heatsink if needed

### 9.3 Mechanical Stress Considerations

Switches are the only components on a PCBA that users physically manipulate. This creates unique reliability requirements.

- **Solder joint strength:** THT joints are 5–10× stronger than SMD for mechanical stress
- **Board support:** Large switches (toggle, rocker) require PCB mounting holes or brackets
- **Actuator alignment:** Switch actuator must align with enclosure button/cutout
- **Travel clearance:** Verify actuator has sufficient travel distance in final assembly
- **ESD:** Some switches generate static when actuated — ground user-accessible metal parts

### 9.4 Conformal Coating & Switches

Conformal coating protects PCBA from moisture and contamination, but switches require special treatment.

- **Tactile switches:** Mask actuator area; coating on contacts prevents proper operation
- **Slide switches:** Mask slider mechanism; coating causes sticking
- **DIP switches:** Mask actuator area; coating on contacts increases resistance
- **Rotary encoders:** Mask shaft and bearing area
- **Membrane switches:** No coating needed — membrane itself is the seal

■ **In-House Best Practice:** For switches in automotive applications (Tesla, Zoox), use THT mounting where possible for superior mechanical strength. If SMD is required, add epoxy staking (EA 0151) to the switch body for additional mechanical retention.

## 10. QUICK REFERENCE SUMMARY

### Tactile Switches:

- **SMD:** 3×4 mm to 6×6 mm; momentary; 100K–1M cycles; consumer electronics
- **THT:** Stronger mechanical attachment; industrial applications
- **Sealed:** IP67 with rubber boot; outdoor/marine use
- **Reflow:** Standard SAC305; verify max temp < 260°C

### Slide Switches:

- **SPST/SPDT/DPST/DPDT:** Maintained contact; 2 or 3 positions
- **Mounting:** DIP, SMD, side-mount, vertical
- **Rating:** 0.3A–3A; signal to low-power switching
- **Orientation:** Verify actuator direction matches silkscreen

### DIP Switches:

- **2–12 positions:** 2.54 mm pitch; SPST per position
- **Use:** Configuration, address, boot mode, feature enable
- **Assembly:** THT; wave or selective solder
- **SMD variant:** 4×4 mm per switch; compact but hard to actuate

### Toggle & Rocker:

- **Toggle:** Lever actuation; panel or PCB mount; 1A–20A
- **Rocker:** Seesaw actuation; illuminated options; 6A–16A
- **PCB mount:** Through-hole with threaded bushing and nut
- **Soldering:** Hand solder; high thermal mass

### Rotary Switches & Encoders:

- **Rotary switch:** 2–12 positions; maintained; channel/range select
- **Encoder:** Continuous rotation; quadrature A/B output
- **With push button:** 5 pins total; click-to-select function
- **Mounting:** Panel mount with PCB support pins

### Membrane Switches:

- **Layers:** Graphic overlay + spacer + circuit + adhesive
- **Types:** Flat (non-tactile), metal dome (tactile), PCB-based
- **Advantages:** Low profile, sealed, customizable, easy to clean
- **Limitations:** No high current, limited life, custom tooling

### Soldering & Process:

- **SMD switches:** Standard reflow; mask from conformal coating

- **THT switches:** Wave, selective, or hand solder
- **Hand solder temp:** 350–400°C; 3–5 sec dwell
- **Mechanical support:** THT preferred for high-stress; epoxy stake SMD if needed
- **Conformal coating:** Mask all actuators and contact areas

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#### References & Standards:

- IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- IPC-A-610: Acceptability of Electronic Assemblies
- IEC 61058: Switches for Appliances
- UL 1054: Standard for Special-Use Switches
- IPC-7527: Stencil Design Guidelines
- IPC-1752A: Materials Declaration Management
- Manufacturer Datasheets: Omron, C&K;, E-Switch, NKK, Grayhill, CTS

*End of Document — Switches in PCBA Assembly Complete Guide*